

Problem Statement

Improper waste segregation is one of the major causes of environmental pollution and inefficient recycling. Many people are unable to identify whether an item belongs to recyclable, biodegradable, or general waste, leading to incorrect disposal. This increases landfill waste, reduces recycling efficiency, and negatively impacts environmental sustainability. An intelligent system that can automatically classify waste can help individuals make better disposal decisions and promote responsible waste management.

Project Overview

EcoSort AI, an AI-powered waste classification system that uses Deep Learning to identify different types of waste from images. The application is built using **TensorFlow**, **MobileNetV2**, and **Streamlit** to provide fast and accurate predictions. Users simply upload an image of waste, and the system classifies it into one of six categories while also providing recycling guidance. The project aims to support sustainability and encourage proper waste segregation through Artificial Intelligence.

Solution Offered

The EcoSort AI system provides:

- AI-powered image classification using the MobileNetV2 deep learning model.
- Classification of waste into six categories: Cardboard, Glass, Metal, Paper, Plastic, and Trash.
- Confidence score for every prediction.
- Recycling recommendations based on the detected waste category.
- A simple and user-friendly Streamlit web application for easy access.

Who Are The End Users?

- Households for proper waste segregation.
- Schools and Colleges for environmental awareness.
- Municipal Corporations for smart waste management.
- Recycling Centers for faster waste sorting.
- Environmental organizations promoting sustainability.
- Students and researchers working on Green AI projects.

Technology Used To Solve The Problem

Artificial Intelligence & Deep Learning

- TensorFlow / Keras: Used to build and train the deep learning model.

- MobileNetV2: Transfer learning model used for accurate image classification.

Programming & Web Application

- Python: Primary programming language.
- Streamlit: Used to develop the interactive web application.

Libraries

- NumPy: Numerical computations.
- Pillow (PIL): Image processing.
- Matplotlib: Training performance visualization.
- Plotly: Interactive data visualization.
- Pandas: Data handling and analysis.

Dataset

- TrashNet Dataset
- Total Images: 2527
- Categories: Cardboard, Glass, Metal, Paper, Plastic, Trash